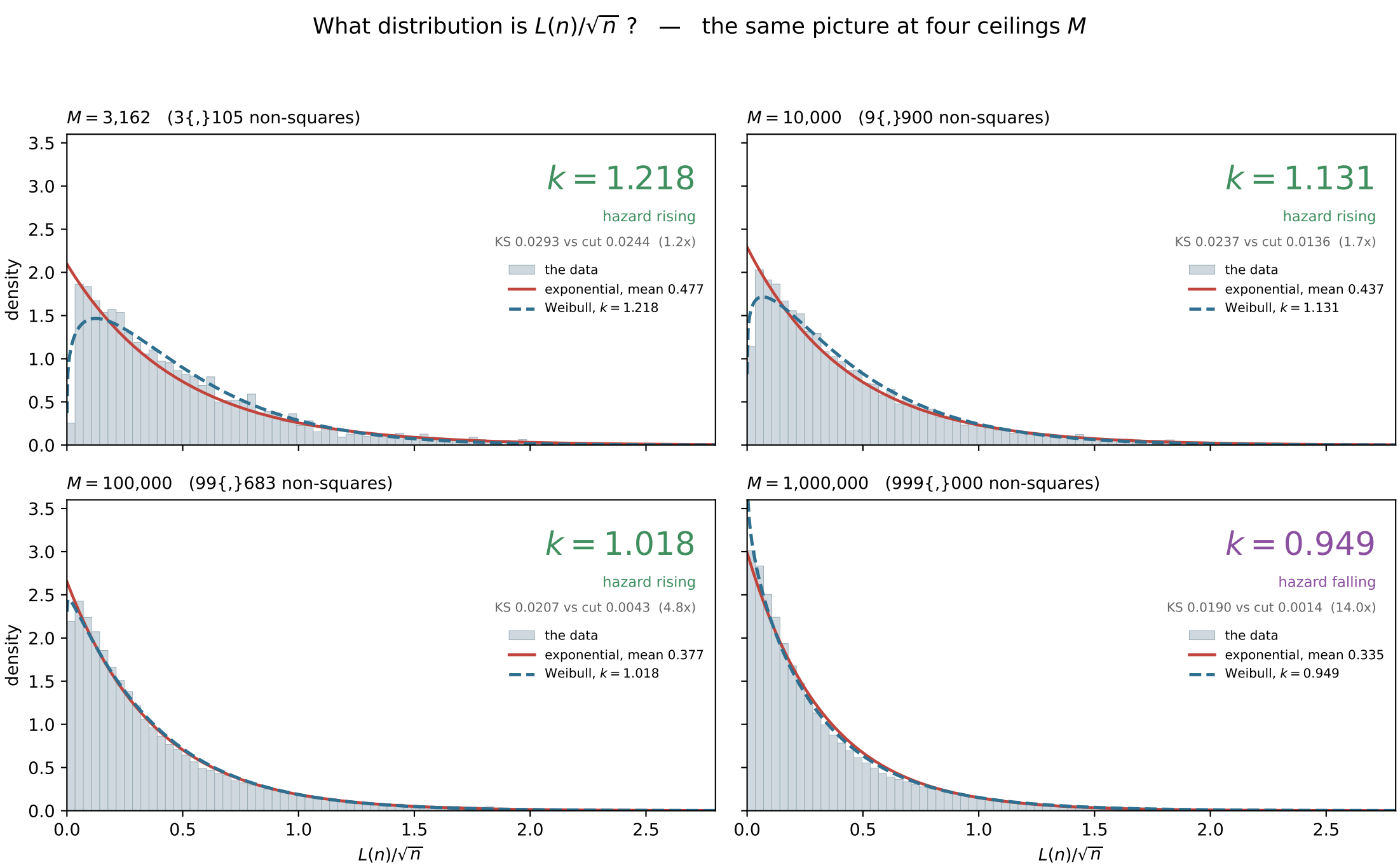
What distribution is $L(n)/\sqrt{n}$? — the same picture at four ceilings M
$M = 3{,}162$ (3{,}105 non-squares)
$k = 1.218$
hazard rising
KS 0.0293 vs cut 0.0244 (1.2x)
the data
exponential, mean 0.477
Weibull, $k = 1.218$
$M = 10{,}000$ (9{,}900 non-squares)
$k = 1.131$
hazard rising
KS 0.0237 vs cut 0.0136 (1.7x)
the data
exponential, mean 0.437
Weibull, $k = 1.131$
$M = 100{,}000$ (99{,}683 non-squares)
$k = 1.018$
hazard rising
KS 0.0207 vs cut 0.0043 (4.8x)
the data
exponential, mean 0.377
Weibull, $k = 1.018$
$M = 1{,}000{,}000$ (999{,}000 non-squares)
$k = 0.949$
hazard falling
KS 0.0190 vs cut 0.0014 (14.0x)
the data
exponential, mean 0.335
Weibull, $k = 0.949$
density
$L(n)/\sqrt{n}$
$k = 1$ is exactly the exponential. The fitted shape falls through it near $M \approx 196{,}000$: green above, purple below.